

Moshe Mash, Ph.D.

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Research Interests

My research centers on environments shared by humans and computer agents: how they interact, coordinate, and reach agreements, and how to design agents that act well alongside people. I combine game theory and mechanism design with machine learning to model, predict, and shape behavior in such settings, from joint exploration and negotiation to coalition formation and fair division. I am developing a new direction that extends this to LLM-based agents: learning what an agent should communicate to a human, and how, from human preference, in mixed human-agent settings such as autonomous and human-driven vehicles.

A related line of work, begun during my postdoc at Carnegie Mellon, studies the people who build AI systems: capturing and modeling the workflows of data scientists as they develop machine learning models, and predicting the moments where they get stuck so that tools can offer targeted support.

A third, applied strand of my research uses large language models and modern NLP for text analysis in specialized domains: authorship attribution, automatic citation generation, and semantic cross-referencing of Hebrew and rabbinic texts (with DICTA), and computational analysis of legal and patent text, including textual studies of disparate outcomes in the patent system. Both involve fine-tuning LLMs for prediction, clustering, and semantic analysis of domain-specific corpora.

Education

Ph.D., Software and Information Systems Engineering | Ben-Gurion University of the Negev | 2014–2018

- **Dissertation:** Reaching Fair Agreements in Group Settings
- **Advisor:** Prof. Ya'akov (Kobi) Gal

M.Sc., Computer Science (AI specialization) | Bar-Ilan University | 2011–2013

- **Thesis:** Joint Exploration with Self-Interested Agents
- **Advisor:** Prof. David Sarne

B.Sc., Software Engineering | Shenkar College of Engineering and Design | 2007–2011

Awards & Recognition

- **Best Paper Award** – ACM Conference on Economics and Computation (2016)
 - **Communications of the ACM Research Highlight** – Which Is the Fairest (Rent Division) of Them All? (2018)
 - **Excellence in Teaching Award** – Ben-Gurion University (2016, 2017)
 - **Ph.D. Scholarship for Academic Excellence** – Ben-Gurion University (2014)
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Academic Positions

Postdoctoral Researcher | Carnegie Mellon University, Robotics Institute | 2019–2022

- Hosted by Prof. Reid Simmons and Dr. Stephanie Rosenthal.
- Conducted research on the behavior and cognitive processes of machine learning practitioners as they develop AI models.
- Designed frameworks to capture, model, and analyze workflows, with emphasis on decision-making, iteration cycles, and knowledge bottlenecks.
- Developed predictive methods to identify critical moments of difficulty (“stuckness”) in ML development, enabling targeted support and improved productivity.
- Collaborated with multidisciplinary teams to integrate human-centric AI approaches into machine learning tools and environments.
- Mentored students in AI and human-computer interaction research.
- Research funded by J.P. Morgan, which renewed the award for a second consecutive year based on the project's strong results.

Research Assistant | Weizmann Institute of Science | 2011–2012

- Developed image-processing algorithms for 3D biomedical images.
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Industry Research Positions

Principal AI Researcher & Founder | MashInnovateAI | 2024–present

- Independent AI research & development consultancy: helps organizations identify where AI can solve their hardest problems, then designs and delivers the solution end to end - applied ML, NLP, LLM fine-tuning, RAG, and agentic and multi-agent systems - from proof of concept to production deployment.
- Clients include technology companies looking to integrate AI into their products and workflows, startups, and universities and academic research groups.

AI Researcher | DICTA – The Israel Center for Text Analysis | 2022–present

- Lead the center's NLP and LLM research on large collections of Hebrew and rabbinic texts: set the research agenda, design the methods, and take systems from prototype to tools in active use by scholars.
- Authorship attribution: built models that predict the author of texts with historically disputed attribution from textual signals alone, reaching high accuracy and resolving long-debated questions of authorship.
- Citation detection and cross-referencing: designed systems that automatically detect, extract, and link citations of classical sources across a large corpus, connecting documents that quote or paraphrase one another.
- Document understanding with LLMs: fine-tuned LLMs and trained custom text embeddings with contrastive learning for domain-specific corpora, powering information extraction, semantic search, text similarity, clustering, and question answering over long, complex documents.
- Linguistic analysis: built deep-learning models for morphological, syntactic, and semantic analysis of long, complex documents at scale.
- Infrastructure and deployment: built the end-to-end training and inference pipeline, with tooling to analyze model outcomes and inspect the training process, and deployed the resulting document-understanding systems to production.

AI Researcher | Diagnostic Robotics | 2019

- Developed machine learning models for the healthcare domain, with applications in risk prediction (e.g., ER diagnoses, hospitalization forecasting).
- Collaborated with clinicians and researchers to refine AI-driven predictive healthcare decision-support systems.

AI Researcher | Amdocs | 2018–2019

- Led the development of a churn-prediction system for at-risk maintenance contract renewals, saving millions of dollars.
 - Designed, built, and deployed the system from scratch, including problem definition, algorithm design, evaluation, and deployment pipeline.
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Teaching & Advising

Ben-Gurion University of the Negev – Software and Information Systems Engineering Dept.

- Teaching Assistant, Databases – Fall-Spring 2016, 2017, 2018
- Teaching Assistant, Analysis and Design of Software Systems – Spring-Summer 2016, 2017, 2018
- Teaching Assistant, Intelligent Systems – Spring-Summer 2015
- Teaching Assistant, Decision Support Systems – Fall-Spring 2015
- Departmental Seminar Coordinator – 2015, 2016, 2017, 2018
- Final Projects Coordinator (4th Year) – 2015, 2016, 2017, 2018
- Final Projects Adviser (Undergraduate) – 2015, 2016

Tel-Aviv Yafo Academic College – Information Systems Dept.

- Lecturer, Databases – Spring-Summer 2015, Fall-Spring 2015
- Lecturer, Game Theory – Spring-Summer 2015
- Teaching Assistant, Databases – Spring-Summer 2014

Bar-Ilan University

- Teaching Assistant, Statistical Methods in Computer Science – Spring-Summer 2014
- Teaching Assistant, Object-Oriented Programming – Fall-Spring 2013, 2014
- Teaching Assistant, Machine Learning – Fall-Spring 2014
- Teaching Assistant, Introduction to Computing – Fall-Spring 2013

Shenkar College of Engineering and Design

- Teaching Assistant, Introduction to Programming Systems – Spring-Summer 2012
- Teaching Assistant, Introduction to Computer Science – Fall-Spring 2012

- Teaching Assistant, Introduction to Object-Oriented Programming – Fall–Spring 2012

College of Management (Academic)

- Teaching Assistant, Object-Oriented Programming – Spring–Summer 2012

High School Teaching (Israel Ministry of Education)

- Mathematics Teacher (2010–2013) – taught Bagrut levels 3–5 points, specializing in students with learning disabilities (via Nitzan Association).
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Professional Service

- **Reviewer:** AAAI, IJCAI, AAMAS, KDD, CHI, EC, ECAI, Group Decision & Negotiation
 - Developed voting algorithm for Best Paper selection (IAAI 2018)
 - Organized departmental seminars at Ben-Gurion University
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Publications

Journal Articles

- Gerhardt D., Marcowitz-Bitton M., Schuster W.M., Elmalech A., Suissa O., Mash M. (2026). Gendered Words and Patent Grant Rates: A Textual Analysis. *Northwestern Journal of Technology and Intellectual Property*, 23(3), Article 2.
- Raphael I, Xiong Z, Sneiderman CT, Raphael RA, Mash M., et al. The T cell receptor landscape of childhood brain tumors. *Science Translational Medicine*. 2025 Mar 19; 17(790): eadp0675.
- Mash, M., Fairstein, R., Bachrach, Y., Gal, K., & Zick, Y. (2020). Human-Computer Coalition Formation in Weighted Voting Games. *ACM Transactions on Intelligent Systems and Technology (TIST)*.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2017). Which Is the Fairest (Rent Division) of Them All? *Journal of the ACM (JACM)*. *Selected as a Communications of the ACM Research Highlight*, 2018.

- Rochlin, I., Sarne, D., & Mash, M. (2014). Joint Search with Self-Interested Agents and the Failure of Cooperation Enhancers. *Artificial Intelligence Journal (AIJ)*.

Conference Proceedings

- Mash, M., Oryol, S., Rosenthal, S., & Simmons, R. (2022). Predicting Data Scientist Stuckness During the Development of Machine Learning Classifiers. *VL/HCC*.
- Mash, M., Rosenthal, S., & Simmons, R. (2021). DSWorkflow: A Framework for Capturing Data Scientists' Workflows. *CHI*.
- Mash, M., Bachrach, Y., Gal, K., & Zick, Y. (2017). How to Form Winning Coalitions in Mixed Human-Computer Settings. *IJCAI*.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2016). Which Is the Fairest (Rent Division) of Them All? *EC*. (Best Paper Award)
- Mash, M., Lin, R., & Sarne, D. (2014). Peer-Design Agents for Reliably Evaluating Distribution of Outcomes in Environments Involving People. *AAMAS*.
- Mash, M., Rochlin, I., & Sarne, D. (2012). Join Me with the Weakest Partner, Please. *IAT*.